Original Article

Study of Sleep Hygiene Index and Sleep Disturbances in Systemic Lupus Erythematosus (SLE) Patients Referred to Rheumatology Clinic in A Hospital in 2024

Yousef Mohammadi Kebar[1] | Aziz Kamran[2] | Elham Nezhadseifi[1,*] | Fatemeh Mirali Ashrafi[1]

[1]*Department of Internal Medicine, School of Medicine, Ardabil University of Medical Sciences, Ardabil, Iran*
[2]*Department of Community Medicine, School of Medicine, Ardabil University of Medical Sciences, Ardabil, Iran*

Use your device to scan and read the article online

Citation Y. Mohammadi Kebar, A. Kamran, E. Nezhadseifi, F, Mirali Ashrafi, **Study of Sleep Hygiene Index and Sleep Disturbances in Systemic Lupus Erythematosus (SLE) Patients Referred to Rheumatology Clinic in A Hospital in 2024.** *Eurasian J. Sci. Technol.*, **2026**, *6*(1), 1-8.

https://doi.org/10.48309/ejst.2026.531148.1254

CC BY

Article info:
Received: 2025-06-28
Accepted: 2025-07-29
Available Online: 2025-08-13
ID: EJST-2506-1254

Keywords:
Systemic Lupus Erythematosus, Sleep Quality, Sleep Hygiene.

ABSTRACT

The chronic and disabling nature of systemic lupus erythematosus (SLE), as well as its own pathogenesis, is associated with the likelihood of developing psychiatric and neurological disorders, such as mood disorders, anxiety, and sleep disturbances. This study aimed to investigate the sleep hygiene index and sleep disturbances in patients with systemic lupus erythematosus (SLE). The statistical population consisted of 50 SLE patients referred to the rheumatology clinic of Ardabil hospital. Data were collected using the Pittsburgh Sleep Quality Index (PSQI) and the Sleep Hygiene Index (SHI) questionnaires and analyzed using descriptive and analytical statistical methods in SPSS version 24. 50 patients with lupus were studied for sleep problems. The mean age of participants was 43.9 ± 12.64 years. Ten patients (20%) had a history of taking sleep medications. 90% were female and 85.4% lived in Ardabil city. Mean BMI: 27.35 ± 5.27. Mean disease duration: 10.58 ± 6.59 years. Mean sleep hygiene score: 7.92 ± 3.91. Mean total sleep score: 5.92 ± 3.52. Male patients had significantly higher sleep disturbance scores than females (11.8 ± 3.7 vs. 7.5 ± 3.72). Sleep hygiene scores were significantly higher in those with a history of sleep medication use (10.5 ± 4.17 vs. 7.28 ± 3.61; P = 0.018). Sleep hygiene scores were significantly lower in patients with hypertension (2.5 ± 1.73 vs. 8.39 ± 3.68; P = 0.003). The sleep quality of patients with systemic lupus erythematosus in Ardabil province is unfavorable. Intervention strategies and health recommendations are needed.

Introduction

Systemic lupus erythematosus (SLE) is an autoimmune disease of unknown origin that affects multiple organs and presents with a wide range of clinical manifestations. The chronic and debilitating nature of SLE, along with its pathogenesis, is associated with an increased likelihood of psychiatric and neurological complications, such as mood disorders, anxiety, and sleep disturbances. Although the prevalence of sleep disorders and factors associated with sleep quality in SLE patients have not yet been

*Corresponding Author: Elham Nezhadseifi: drelham320@yahoo.com

clearly defined, studies suggest that such disorders are highly common—affecting more than 80% of this population. Sleep disturbances significantly exacerbate systemic lupus erythematosus (SLE) management through multiple pathways: (1) deteriorating patients' quality of life, (2) amplifying fatigue and depressive symptoms, (3) contributing to work absenteeism, and (4) escalating healthcare expenditures [1]. Sleep is a fundamental biological process, likely linked to energy preservation. Difficulties in initiating sleep, maintaining sleep, and/or early morning awakening are all considered components of sleep disorders [2]. Poor or insufficient sleep can exacerbate fatigue—a common complaint in SLE patients—and negatively impact health-related quality of life (HRQoL) and cognitive function [3]. Therefore, assessing sleep quality should be regarded as an essential part of the comprehensive evaluation of SLE patients, as it provides valuable insights for optimal disease management and treatment decision-making.

Moreover, developing strategies to improve sleep quality in SLE patients underscores the ongoing importance of evaluating mental health. Recently, sleep hygiene education has demonstrated potential as an effective intervention strategy to address the growing public health concerns related to sleep complaints in the general population. However, consistent research investigating the application and clinical significance of sleep hygiene components remains lacking [4].

The aim of this study was to evaluate the sleep hygiene index and sleep disturbances in patients with systemic lupus erythematosus (SLE) referred to the rheumatology clinic of Imam Khomeini Hospital in Ardabil.

Methods

Study Design and Participants

This descriptive cross-sectional study was conducted on 50 patients diagnosed with systemic lupus erythematosus (SLE) who were referred to the rheumatology clinic of Hospital in Ardabil city at year 2024. Data were collected using the Pittsburgh Sleep Quality Index (PSQI) and the Sleep Hygiene Index (SHI), in accordance with the study's research objectives.

Pittsburgh Sleep Quality Index (PSQI):

This questionnaire consists of 19 items rated on a 4-point Likert scale ranging from 0 to 3. It includes seven subscales:

1. Subjective Sleep Quality
2. Sleep Latency
3. Sleep Duration
4. Habitual Sleep Efficiency
5. Sleep Disturbances
6. Use of Sleep Medications
7. Daytime Dysfunction

The PSQI was originally developed by Buysse *et al*. in 1989, and its internal consistency reliability (Cronbach's alpha) was reported at 0.83. The Persian version of the questionnaire has shown acceptable validity (0.86) and reliability (0.89). The total PSQI score ranges from 0 to 21, with each subscale scoring between 0 and 3. Interpretation of scores is as follows:

- 0: No sleep problem
- 1: Mild sleep problem
- 2: Moderate sleep problem
- 3: Severe sleep problem

A global score greater than 5 indicates poor sleep quality or unfavorable.

Sleep Hygiene Index (SHI): The SHI is a 13-item self-report measure designed by Mastin et al. in 2006 to assess environmental and behavioral factors that can adversely affect sleep. Each item is scored on a five-point scale (Always, Frequently, Sometimes, Rarely, Never), and the total score ranges from 0 to 52. Higher scores indicate poorer sleep hygiene. The SHI has demonstrated acceptable internal consistency (Cronbach's alpha = 0.66) and test–retest reliability (0.71). A positive correlation (r = 0.48) has also been reported between SHI and the PSQI. The validity and reliability of the SHI have also been confirmed in the Iranian population.

Data Analysis: Data were analyzed using IBM SPSS Statistics software, version 26. Quantitative variables were expressed as means and standard deviations, and qualitative variables were reported as frequencies and percentages. The Chi-square test was used to assess sleep quality across qualitative variables, and one-way ANOVA was applied to compare

sleep hygiene scores among different groups. A p-value of less than 0.05 was considered statistically significant.

Ethical Considerations: The study was conducted following approval from the university's Ethics Committee (Ethics Code: IR.ARUMS.MEDICINE.REC.1402.037).

Results

In this study, 50 patients with systemic lupus erythematosus (SLE) were evaluated for sleep quality and sleep hygiene. The average age of participants was 43.9 ± 12.64 years, with a mean BMI of 27.35 ± 5.27, and a mean disease duration of 10.58 ± 6.59 years. most patients were female (90%) and resided in Ardabil city (85.4%). (Table 1) According to the Pittsburgh Sleep Quality Index (PSQI), 60% of patients had poor sleep quality, and 82% showed moderate sleep disturbances. Most of clinical variables in patients was Antiphospholipid syndrome (APS). (Figure 1) Male patients had a significantly higher mean PSQI score (11.8 ± 3.7) compared to females (7.5 ± 3.72). Patients with a history of sleep medication use demonstrated significantly worse sleep hygiene scores (10.5 ± 4.17) compared to non-users (7.28 ± 3.61) (P = 0.018). (Table 2) Additionally, patients with hypertension had significantly lower sleep hygiene scores (2.5 ± 1.73) compared to those without hypertension (8.39 ± 3.68) (P = 0.003). The use of sleep medications was also significantly more common in the group with poor sleep quality (P = 0.003), while no significant differences were observed regarding place of residence, diabetes, or educational status (Table 3).

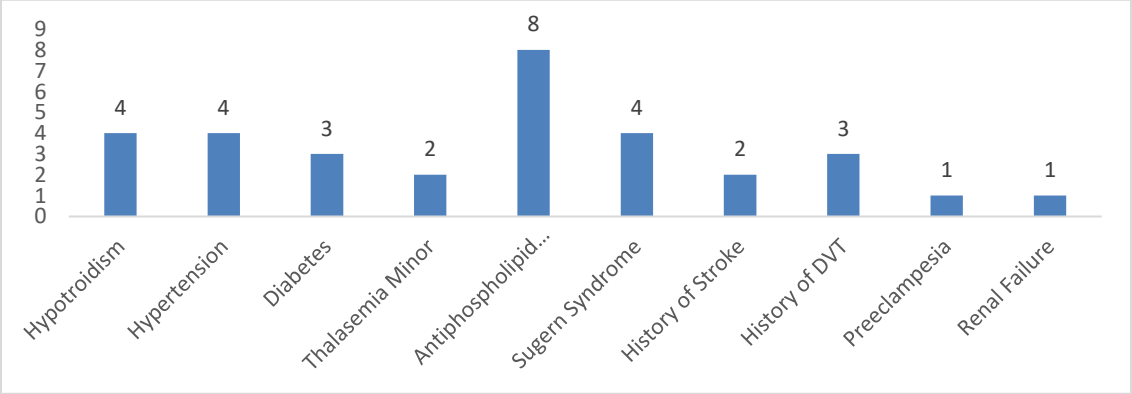


Figure 1 Frequency of clinical variables in study participants

Table 1 demographic variables of participants

Variables	levels	n	%
Sex	Male	5	10
	Female	45	90
Education	Illiterate	4	8
	Primary	6	12
	High school	8	16
	Diploma	15	30
	University	17	34
Job	Housekeeper	36	72
	Students	2	4
	Teacher	4	8
	Free job	6	12
	Employee	1	2
	retired	1	2
Residence place	Ardabil	41	4/85
	Other cities	7	6/14
Use of sedation	+	10	20
	-	40	80

Table 2 Mean of sleep health score in SLE patients by variables

Variables	levels	Mean	SD	p-value
Sex	Male	11.8	3.7	0.018
	Female	7.5	3.72	
Education	Illiterate	7	4.4	0.63
	Primary	9.3	3.61	
	High school	6.63	4.63	
	Diploma	8.73	3.77	
	University	7.53	3.83	
Job	Housekeeper	7.64	3.99	0.25
	Students	6	2.83	
	Teacher	8.5	0.58	
	Free job	9.5	4.04	
	Employee	3	1	
	retired	15	1	
Residence place	Ardabil	7.98	4.04	0.74
	Other cities	7.43	3.87	
Use of sedation	+	10.5	4.17	0.018
	-	7.28	3.61	
HTN	+	2.5	1.73	0.003
	-	8.39	3.68	
DM	+	7.67	3.06	0.91
	-	7.94	3.98	
Sleep disorders	None	5.75	1.71	0.51
	Moderate	8.03	3.94	
	severe	8.4	4.93	
Sleep quality	Good	5.6	2.93	0.001
	Bad	9.47	3.74	

Table 3 The mean sleep health score among SLE patients, stratified by sleep quality according to PSQI

Variables	levels	Sleep Quality				p-value
		Favorable		Unfavorable		
		n	%	n	%	
Sex	Male	1	5	4	13.3	0.63
	Female	19	95	26	86.7	
Education	Illiterate	2	10	2	6.7	0.21
	Primary	0	0	6	20	
	High school	5	25	3	10	
	Diploma	6	30	9	30	
	University	7	35	10	33.3	
Job	Housekeeper	15	75	21	70	0.16
	Students	2	10	0	0	
	Teacher	0	0	4	13.3	
	Free job	2	10	4	13.3	
	Employee	1	5	0	0	
	retired	0	0	1	3.3	
Residence place	Ardabil	15	75	26	92.9	0.11
	Other cities	5	25	2	7.1	
Use of sedation	+	0	0	10	33.3	0.003
	-	20	100	20	66.7	
HTN	+	2	10	2	6.7	1
	-	18	90	28	93.3	
DM	+	0	0	3	10	0.26
	-	20	100	27	90	

Discussion

This study is the first in Northwestern Iran to assess sleep characteristics in patients with systemic lupus erythematosus (SLE). Compared to healthy Iranian individuals, SLE patients scored higher in all dimensions and total scores of the Pittsburgh Sleep Quality Index (PSQI), indicating poorer sleep quality. Particularly, subjective assessments revealed major issues in sleep latency and sleep duration [5]. For instance, the mean score for sleep latency (1.64) was significantly higher than values reported in studies by Zhao (0.47) and Wu (0.49), and closer to Mirbagher's Iranian study (1.45), suggesting worse sleep initiation among Iranian SLE patients [6-8]. Similarly, in other PSQI components like sleep duration and disturbances, patients in this study showed poorer outcomes, though scores for daytime dysfunction and sleep efficiency were comparatively better, possibly indicating resilience in daily performance despite poor sleep. The global PSQI score was 5.92, higher than Zhao's (3.59) and Wu's (3.45), but lower than Mirbagher's (7.06), Khan's (7.1), and Ozer's (7.81), pointing to moderate sleep impairment among Ardabil patients. Sleep disturbances, known to affect metabolism, tissue repair, and systemic homeostasis, are recognized contributors to disease severity and management complexity in SLE [8,9].

Sleep is an active biological process through which metabolism, tissue repair, and overall homeostatic balance are maintained. Sleep disturbances have emerged as significant determinants of disease course and severity across many conditions, including rheumatic disorders such as systemic lupus erythematosus (SLE) [10]. Poor sleep quality has been linked to depressive mood, disease activity, prednisone use, lack of physical activity, and elevated levels of inflammatory cytokines, all contributing to an overall decline in sleep quality [11]. Subjective studies have shown that up to 80% of SLE patients experience poor sleep quality [12]. Three studies comprehensively examine the complex relationships between sleep, mental health, and neural structure. Freeman et al. (2020) establish that sleep disturbances - including insomnia and nightmares - function dually as both symptoms of psychiatric conditions (e.g., depression, anxiety) and independent risk factors for their onset or worsening. This evidence underscores the critical role of sleep-focused interventions within comprehensive psychiatric care. Van Dyk *et al.* (2019) focus on adolescents with insomnia, revealing a significant association between poor sleep quality, inadequate sleep hygiene, and increased psychiatric symptoms such as depression and anxiety, underscoring the need for early sleep interventions in this vulnerable age group. Meanwhile, Santos *et al*. (2023) explore the link between sleep patterns and cortical gyrification changes, demonstrating the critical role of adequate sleep in optimal brain development and organization during adolescence and young adulthood, while warning that sleep deprivation may negatively impact neurodevelopment and cognitive-emotional functioning [13-15].

Collectively, these findings support an integrated framework in which improving sleep through interventions such as sleep hygiene education, cognitive-behavioral therapy for insomnia (CBT-I), and sleep disorder monitoring in clinical populations can serve as an effective strategy for enhancing mental health and neural functioning, particularly in high-risk groups like adolescents. Collectively, these studies highlight two critical imperatives: (1) the integration of sleep evaluation and therapeutic interventions into standard mental health protocols, and (2) expanded investigation into the neurobiological pathways mediating sleep-psychopathology interactions. Patients often report poor sleep, frequent awakenings, and restlessness. These issues can impair daytime functioning due to persistent fatigue and sleepiness, which may reflect profound physical exhaustion, decreased energy, or even mood disorders such as depression. Poor or insufficient sleep can exacerbate fatigue, the most commonly reported symptom among SLE patients, and may disrupt health-related quality of life (HRQoL) and cognitive performance. Sleep disturbances are also recognized as one of the core symptoms of depression, with associated issues like impaired concentration and lethargy potentially manifesting as fatigue [16]. Therefore,

improving sleep quality may also lead to enhanced overall functioning in patients with SLE.

Sleep problems are frequently reported as one of the less-recognized needs of patients with systemic lupus erythematosus (SLE) [17]. Sleep quality in this study was assessed using the Pittsburgh Sleep Quality Index (PSQI), a self-report questionnaire with a possible score range from 0 to21, where higher scores indicate poorer sleep quality. This instrument has demonstrated strong psychometric properties, with a diagnostic sensitivity of 89.6% and specificity of 86.5%, and has shown a good correlation with the Sleep Hygiene Index [18]. Sleep complaints constitute a significant part of the symptomatology of many rheumatic diseases, including SLE [19]. Current evidence suggests sleep disorders exhibit high prevalence among SLE patients, likely impacting over 50% of individuals. However, existing studies in this domain have been constrained by notable methodological shortcomings. In SLE patients, sleep disturbances—particularly poor sleep quality—have been positively correlated with disease activity and various clinical and demographic factors [20]. In this study, male patients and those with a history of sleep medication use exhibited poorer sleep hygiene. Moreover, the use of sleep medications was significantly associated with poor overall sleep quality. Interestingly, even non-hypertensive patients showed poor sleep hygiene, although only four hypertensive patients were included in the sample. Given the relatively high prevalence of hypertension in the general population, further studies with larger samples are needed to better interpret these findings. Other studies have identified additional risk factors; for instance, Wu [7] reported that residential location and longer disease duration were significantly associated with higher PSQI scores. In Ozer's study [9], comorbid conditions and disease duration correlated with increased sleep problems, and in Mirbagher's study [8], older age and higher BMI were directly associated with greater sleep disturbances in SLE patients. In conclusion, comprehensive investigations remain imperative to elucidate the pathophysiological mechanisms underlying sleep disturbances in SLE patients. Given the questionnaire-based nature of the current study, a key limitation lies in the subjective nature of the data, which calls for caution when interpreting the results.

Conclusion

The results showed that the sleep quality of patients with systemic lupus erythematosus (SLE) in Ardabil province is unfavorable. Therefore, appropriate health interventions and recommendations are necessary in this regard. It is also suggested that further cohort and prospective studies with control groups be conducted. Additionally, studies should include patients with SLE from other academic centers to examine possible differences compared to patients in Ardabil. Additionally, we recommend implementing questionnaire-based studies with larger sample sizes focusing on hypertensive SLE patients to more comprehensively examine the association between hypertension and sleep hygiene in this population.

Acknowledgments

We thank all the participants in the study.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Ethics Approval

The research protocol received approval from the ethical committee of Ardabil University of Medical Sciences (IR.ARUMS.MEDICINE.REC.1402.037).

Conflict of Interests

The authors declare that they have no conflict of interest.

Authors' Contribution

All authors participated in the planning and execution of the study, the analysis of findings, and the drafting of the manuscript.

Data Availability

The corresponding author's data supporting this study's findings are available upon reasonable request.

Consent for Publication

Not applicable.

References

[1] Sahebari, M., Ravanshad, S., Ravanshad, Y., Rezaeitalab, F., Bayegi, H.R.P., Asadpour, H., Rezaieyazdi, Z. A survey on sleep disorders and related hormones in patients with newly diagnosed systemic lupus erythematosus. *Mediterranean Journal of Rheumatology*, **2021**, 32(2), 148-157.

[2] Chiu, H.Y., Wang, M.Y., Chang, C.K., Chen, C.M., Chou, K.R., Tsai, J.C., Tsai, P.S. Early morning awakening and nonrestorative sleep are associated with increased minor non-fatal accidents during work and leisure time. *Accident Analysis & Prevention*, **2014**, 71, 10-14.

[3] Kasitanon, N., Achsavalertsak, U., Maneeton, B., Wangkaew, S., Puntana, S., Sukitawut, W., Louthrenoo, W. Associated factors and psychotherapy on sleep disturbances in systemic lupus erythematosus. *Lupus*, **2013**, 22(13), 1353-1360.

[4] Yin, R., Li, L., Xu, L., Sui, W., Niu, M.E., Xu, R., Srirat, C. Association between depression and sleep quality in patients with systemic lupus erythematosus: a systematic review and meta-analysis. *Sleep and Breathing*, **2022**, 26(1), 429-441.

[5] Hamzeh, S., Safari-Faramani, R., Khatony, A. Effects of aromatherapy with lavender and peppermint essential oils on the sleep quality of cancer patients: a randomized controlled trial. *Evidence-Based Complementary and Alternative Medicine*, **2020**, 2020(1), 7480204.

[6] Zhao, Q., Deng, N., Chen, S., Cui, Y., Du, X., Gu, Z. Systemic lupus erythematosus is associated with negatively variable impacts on domains of sleep disturbances: a systematic review and meta-analysis. *Psychology, Health & Medicine*, **2018**, 23(6), 685-697.

[7] Wu, L., Shi, P.L., Tao, S.S., Tao, J.H., Wu, G.C. Decreased sleep quality in patients with systemic lupus erythematosus: a meta-analysis. *Clinical Rheumatology*, **2021**, 40(3), 913-922.

[8] Khan, A., Pakozdi, A., Rajakariar, R., Cove-Smith, A., Lewis, M., Pyne, D. E090 Sleep disturbances in lupus patients. *Rheumatology*, **2019**, 58(Supplement_3), kez110. 088.

[9] Özer, S., Kankaya, H., Gün, R., Yeler, N., Marangoz, Ö., Bozça, H. Factors affecting sleep quality in patients with systemic lupus erythematosus. *Lupus*, **2022**, 31(1), 39-44.

[10] Chung, W., Lin, C., Kao, C. Association of systemic lupus erythematosus and sleep disorders: a nationwide population-based cohort study. *Lupus*, **2016**, 25(4), 382-388.

[11] Costa, D.D., Bernatsky, S., Dritsa, M., Clarke, A.E., Dasgupta, K., Keshani, A., Pineau, C. Determinants of sleep quality in women with systemic lupus erythematosus. *Arthritis Care & Research: Official Journal of the American College of Rheumatology*, **2005**, 53(2), 272-278.

[12] Rady, H.M., Ghanim, D.H. Sleep disturbance in female patients with systemic lupus erythematosus and its relation to disease parameters. *The Egyptian Rheumatologist*, **2013**, 35(3), 127-132.

[13] Freeman, D., Sheaves, B., Waite, F., Harvey, A., Harrison, P. Sleep disturbance and psychiatric disorders: the non-specific as essential in understanding and treating mental ill health. *Lancet Psychiatry*, **2020**, 7(7), 628-637.

[14] Van Dyk, T.R., Becker, S.P., Byars, K.C. Rates of mental health symptoms and associations with self-reported sleep quality and sleep hygiene in adolescents presenting for insomnia treatment. *Journal of Clinical Sleep Medicine*, **2019**, 15(10), 1433-1442.

[15] Lima Santos, J.P., Hayes, R., Franzen, P.L., Goldstein, T.R., Hasler, B.P., Buysse, D.J., Ladouceur, C.D. The association between cortical gyrification and sleep in adolescents and young adults. *Sleep*, **2024**, 47(1), zsad282.

[16] Iaboni, A., Ibanez, D., Gladman, D.D., Urowitz, M.B., Moldofsky, H. Fatigue in systemic lupus erythematosus: contributions of disordered sleep, sleepiness, and depression. *The Journal of Rheumatology*, **2006**, 33(12), 2453-2457.

[17] Danoff-Burg, S., Friedberg, F. Unmet needs of patients with systemic lupus erythematosus. *Behavioral Medicine*, **2009**, 35(1), 5-13.

[18] Buysse, D. The Pittsburgh Sleep Quality Index: a new instrument for psychiatric practice and

research. *Psychiatry*, **2003**, 18, 498.

[19] Palagini, L., Tani, C., Bruno, R., Gemignani, A., Mauri, M., Bombardieri, S., Mosca, M. Poor sleep quality in systemic lupus erythematosus: does it depend on depressive symptoms? *Lupus*, **2014**, 23(13), 1350-1357.

[20] Kok, V.C., Horng, J.T., Hung, G.D., Xu, J.L., Hung, T.W., Chen, Y.C., Chen, C.L. Risk of autoimmune disease in adults with chronic insomnia requiring sleep-inducing pills: a population-based longitudinal study. *Journal of General Internal Medicine*, **2016**, 31(9), 1019-1026.

Copyright © 2026 by SPC (Sami Publishing Company) + is an open access article distributed under the Creative Commons Attribution License (CC BY), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.